Global nonvanishing after lower-period cancellation at weight five

Abstract

Two explicit third-order recurrence certificates establish the adjacent coefficient signs needed to cancel $\zeta(3)$ at every even index. A uniform recurrence cone proves positivity from three initial values; the argument does not extrapolate finite sign checks. The certificates are checked independently in $\mathbb{Z}[n,t]$. Lean verifies the cone induction, its coefficient inequalities, and positivity of the specified recurrence states. Their identification with the Laurent coefficients is a paper proof here. The winning primitive arithmetic bound, and hence irrationality of $\zeta(5)$, remain unproved.

1 The coefficients and the claim

For $a = 0, 1$ and $n \geq 2$, put

$$R_n^{(a)}(t) = (n!)^3 \frac{\prod_{v=1}^{n-a}(t-v)(t+n+v)}{\prod_{j=0}^{n}(t+j)^5}.$$

Let ϕ_3 sum all third-order partial-fraction coefficients in the integer pole orbit, and set $v_n^{(a)} = \phi_3(R_n^{(a)})$. In the original normalization,

$$T_{n,m}(t) = \frac{(n!)^5}{(m!)^2} \frac{\prod_{v=1}^{m}(t-v)(t+n+v)}{\prod_{j=0}^{n}(t+j)^5}, \qquad \alpha_{n,n} = v_n^{(0)}, \quad \alpha_{n,n-1} = n^2 v_n^{(1)}.$$

For even n, $\alpha_{n,m}$ is the coefficient of $\zeta(3)$ in $\mathcal{A}_{n,m} := \sum_{k\geq 1} T_{n,m}(k)$. The sums converge absolutely by rational decay and are strictly positive: their summands vanish for $k \leq m$ and are positive for $k > m$. Partial-fraction summation identifies α with the stated coefficient. Reflection $T_{n,m}(-n-t) = -T_{n,m}(t)$ cancels the even zeta terms; proper decay gives $\sum_j c_{j,1} = 0$, leaving a rational constant and only $\zeta(3), \zeta(5)$.

Theorem 1. *For every even $n \geq 2$,*

$$(-1)^{n/2}\alpha_{n,n} > 0, \qquad (-1)^{n/2+1}\alpha_{n,n-1} > 0.$$

In particular $\alpha_{n,n}\alpha_{n,n-1} < 0$. Consequently

$$(-1)^{n/2+1}[\alpha_{n,n-1}\mathcal{A}_{n,n} - \alpha_{n,n}\mathcal{A}_{n,n-1}] > 0.$$

The last expression belongs to $\mathbb{Q} + \mathbb{Q}\zeta(5)$.

The proof is computer assisted through finite polynomial certificates. The two accompanying JSON files specify all polynomials below by integer coefficient lists in ascending powers; the standard-library verifier checks the displayed identities by complete coefficient expansion.

2 An exact index recurrence

Fix a. The certificate gives $q_0, \dots, q_3$ and $X(n,t) = \sum_{k=0}^{30} x_k(n) t^k$. Define

$$U = t^6(t+2n-a+1), \qquad V = (t-n+a-1)(t+n+6)^6,$$

$$H_\ell(n,t) = \prod_{i=1}^{2\ell}(t-n+a-i) \prod_{i=1}^{4\ell}(t+2n-a+i) \prod_{i=2\ell+1}^{6}(t+n+i)^6.$$

The first checked identity, in $\mathbb{Z}[n,t]$, is

$$U(n,t)X(n,t+1) - V(n,t)X(n,t) = \sum_{\ell=0}^{3} q_\ell(n) H_\ell(n,t). \tag{1}$$

The files also specify $P_0, \dots, P_3$ and verify

$$q_\ell(n) \prod_{i=2\ell+1}^{6}(n+i)^3 = g_a(n) P_\ell(n), \qquad g_0 = \prod_{i=3}^{6}(n+i)^3, \quad g_1 = \prod_{i=2}^{6}(n+i)^3.$$

All P_ℓ have degree 25 for $a=0$ and degree 36 for $a=1$.

To interpret the certificate, put

$$D(n,t) = \prod_{i=1}^{6}(t+n+i)^6, \quad B(n,t) = \frac{R_n^{(a)}(t)}{D(n,t)}, \quad G(n,t) = V(n,t)X(n,t)B(n,t).$$

Direct cancellation gives the rational-function identities

$$\frac{B(n,t+1)}{B(n,t)} = \frac{t^6(t+2n-a+1)}{(t-n+a)(t+n+7)^6}, \qquad B(n,t)H_\ell(n,t) = \frac{R_{n+2\ell}^{(a)}(t)}{\prod_{i=1}^{2\ell}(n+i)^3}.$$

Thus (1), multiplied by B, gives a rational telescoper whose right side is a linear combination of the four shifted kernels.

The functional ϕ_3 is translation invariant: replacing $G(t)$ by $G(t+1)$ shifts each integer pole by one without changing its third-order coefficient. All these rational functions have finitely many integer poles. Applying ϕ_3, clearing the factorial factors, and using $g_a(n) > 0$ therefore proves

$$\sum_{\ell=0}^{3} P_\ell(n) v_{n+2\ell}^{(a)} = 0 \qquad (n \geq 2). \tag{2}$$

This does not assume numerical linear independence of zeta values: the functional is applied before evaluating any period.

3 A cone that preserves the sign

The verifier checks that each coefficient of

$$-P_0(x+2), \quad P_1(x+2), \quad P_2(x+2), \quad P_3(x+2)$$

is a nonnegative integer, with a positive constant term. The same is true of

$$\Omega(x) = 10^6 P_2(x+2) - 10^9 P_3(x+2) - 1000 P_1(x+2) + P_0(x+2).$$

These are polynomial positivity certificates for all real $x \geq 0$, not sign evaluations at finitely many indices. For $n \geq 2$, set

$$A = -P_0(n) > 0, \quad B = P_1(n) > 0, \quad C = P_2(n) > 0, \quad D = P_3(n) > 0.$$

Then

$$10^6 C > 10^9 D + 1000B + A. \tag{3}$$

Let $u_k^{(a)} = (-1)^{k+1+a} v_{2k+2}^{(a)}$. Equation (2) becomes

$$Du_{k+3} = Cu_{k+2} - Bu_{k+1} - Au_k,$$

where the coefficients are evaluated at $n = 2k+2$. Suppose $u_k > 0$, $u_{k+1} \geq 1000u_k$, and $u_{k+2} \geq 1000u_{k+1}$. Then (3) gives

$$Du_{k+3} \geq \left(C - \frac{B}{1000} - \frac{A}{10^6}\right) u_{k+2} \geq 1000 D u_{k+2}.$$

The cone is therefore preserved.

Direct Laurent expansion at the first three indices gives

$$(u_0^{(0)}, u_1^{(0)}, u_2^{(0)}) = \left(882, \frac{9149875}{2}, \frac{225341478859}{5}\right),$$

$$(u_0^{(1)}, u_1^{(1)}, u_2^{(1)}) = \left(\frac{785}{8}, \frac{16148335}{128}, \frac{89419424171}{162}\right).$$

Both triples satisfy the cone by exact rational comparisons. Induction proves $u_k^{(0)}, u_k^{(1)} > 0$ for every $k \geq 0$, establishing the two asserted coefficient signs. The cancellation weights $\alpha_{n,n-1}$ and $-\alpha_{n,n}$ have the common sign $(-1)^{n/2+1}$; strict positivity of both $\mathcal{A}$ sums proves the last inequality in the theorem. The $\zeta(3)$ coefficient cancels algebraically. No separate all-index assertion about the nonzero $\zeta(5)$ coefficient is needed or made.

4 Verification scope and remaining arithmetic

The files `pade-sign-recurrence-0.json` and `pade-sign-recurrence-1.json` contain the complete certificates. Run

```
python scripts/verify_sign_recurrences.py
```

from the included Lean project. This verifier uses only integer and rational arithmetic from Python's standard library. SymPy 1.14.0 is needed only to rediscover the certificates, not to check them.

The Lean module `PadeSignRecurrence.lean` proves the cone step, its induction, the coefficient inequalities using nonnegative Horner coefficient lists, and positivity of the two explicitly defined recurrence states. The step identifying those states with the actual Laurent-coefficient sequences uses the paper argument from (1) to (2). That connection is not yet formalized in Lean; no complete Lean theorem about the global Laurent-coefficient signs is claimed here.

Nonvanishing after lower-period cancellation is now established for this family at every even index. The primitive arithmetic multiplier remains

$$K_n = \frac{d_n^5 \sum_j |w_{n,j}|}{h_n}, \qquad d_n = \operatorname{lcm}(1, \ldots, n),$$

where the $w_{n,j}$ are integer cancellation weights obtained by clearing the two rational weights, and h_n is the full coefficient gcd. The needed limiting comparison between K_n and the analytic decay is still unproved. This note does not prove irrationality of $\zeta(5)$ or of all odd zeta values.